Navodit Maheshwari

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Education

Masters of Science Natural Language Processing, UC Santa Cruz — Sept 2025 – Jan 2027

- GPA: 4.0/4.0
- **Coursework:**Deep Learning for Natural Language Processing, Data Science and Machine Learning Fundamentals, Natural Language Processing I

Bachelors of Science in Computer Science, UC Santa Cruz — Sept 2022 – June 2025

- GPA: 3.5/4.0
- **Coursework:** Artificial Intelligence, Intro to Natural Language Processing, Machine Learning, Principles of Computer Systems Design, Computer Architecture, Introduction to Probability Theory

Skills

Programming Languages: Python, Java, C, C++, JavaScript, TypeScript, SQL, Go, RISC-V Assembly, HTML, CSS
Frameworks & Libraries React, Flask, Spring Boot, Tailwind CSS, Node.js (basic), PyTorch, TensorFlow
Tools & Platforms: Git, Firebase (incl. Firestore & Authentication), UNIX/Linux, macOS, Windows, npm, Postman, Chrome DevTools, VS Code, Docker, Firestore, MySQL, AutoCAD, Agentic AI
Software Engineering: Data Structures & Algorithms, Object-Oriented Design, REST APIs, System Design Basics

Experience

Graduate Researcher – LLM Steering & Alignment, Lab of Chenguang Wang — Feb 2026 – Present

- Identified activation entanglement in single-layer DIM steering vectors, showing consistent out-of-distribution degradation across three LLM families.
- Developed a multi-layer steering approach that redistributes high-magnitude activations across transformer layers, reducing unintended side effects.
- Used GemmaScope and SAELens to analyze representation quality through reconstruction error, sparsity (L0), and entropy-based metrics.
- Evaluated steering effectiveness and safety across 17 datasets on Gemma 3 4B, Llama 3.1 8B, and Qwen 3 4B.

Tutor for Engineering Principles of Electronics, Baskin Engineering – Santa Cruz, CA — Jan 2026 – Present

- Supported over 200 students in a cross-disciplinary course covering sensors, analog/digital electronics, and communication systems.
- Provided detailed, rubric-aligned feedback on assignments to help students improve quickly and understand core concepts.
- Answered student questions on course material, assignments, and logistics with clear and timely communication.

Projects

TOAD (To Outline A Destination) — Mar 2025

- Led end-to-end development of a multi-user web application from product concept to deployment.
- Designed a scalable real-time data model using Firestore to support concurrent user collaboration.
- Built a drag-and-drop itinerary system enabling intuitive reordering and cross-user synchronization.
- Partnered usability-driven UI decisions with backend architecture to ensure reliability and performance.

Banana Bulletin — Jan 2024

- Led a team of four to build a browser extension using HTML and JavaScript that analyzes news article titles and recommends related articles from different publications.
- Guided feature decisions and task division, keeping the team focused and on schedule during the hackathon.
- Designed and implemented the recommendation logic to surface multiple perspectives on the same topic.
- Presented the project to judges and won the hackathon by demonstrating a practical tool for comparing news coverage and reducing information bias.